Water Draining from a Bottle

Aim

The aim of this experiment is to study how water drains from a small hole in a bottle and to test Torricelli's law.

Theory

According to Torricelli's law, the speed of water leaving a hole is

$$v = \sqrt{2gh},$$

where v is the exit speed, g is gravitational acceleration, and h is the height of water above the hole.

If the bottle has cross-sectional area A and the hole has area a, then

$$A\frac{dh}{dt} = -a\sqrt{2gh}.$$

Solving this gives

$$\sqrt{h(t)} = \sqrt{h_0} - kt,$$

where k is a constant. Therefore, $\sqrt{h}$ should decrease approximately linearly with time.

Materials

Plastic bottle, water, ruler, marker, phone camera or stopwatch, small pin or nail, and a bowl to collect water.

Procedure

A small hole was made near the bottom of the bottle. Height marks were drawn on the bottle starting from the hole. The bottle was filled with water, and the water height was recorded at regular time intervals as it drained.

Expected Result

The water level should fall quickly at first and more slowly later. The graph of h against time should be curved, while the graph of $\sqrt{h}$ against time should be approximately a straight line.

Sources of Error

Possible errors include inaccurate height readings, an irregular hole shape, changing hole size, turbulent flow, and the bottle not having a constant cross-sectional area.

Conclusion

The experiment demonstrates that water drains faster when the water level is high and slower when the water level is low. If the graph of $\sqrt{h}$ versus time is approximately linear, the result supports Torricelli's law.